

FRANK WAN

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PROFESSIONAL EXPERIENCE

Amazon Games Studio

Seattle, Washington

Software Engineer

May - October 2025

- Engineered a seamless integration between an Unreal Engine 5 Norse-themed third-person adventure game and Amazon's generative AI platform (Ermis), enabling fully dynamic, context-aware NPC dialogue and interactions.
- Collaborated with ML science and gameplay teams to refine AI-driven companion behavior where NPCs like Thor, Freya, Loki, and Ratatoskr dynamically respond to player commands and collaborate in combat and puzzle-solving.
- Reduced Ermis NPC dialogue latency from 3s to less than 0.5s by replacing audio-based communication with a text-based system, improving stability and responsiveness.

Amazon - Project Kuiper/LEO

Redmond, Washington

Software Engineer

2023 - 2025, 2026 - Present

- Led the development effort for Initial Network Entry (INE), the process for customer terminals (ground-based satellite antennas) to establish first-time connectivity with Amazon's low Earth orbit satellite broadband network.
- Brought up multiple hardware versions of consumer and enterprise customer terminals, achieving uplink and downlink connectivity with lab and in-orbit satellites.
- Investigated, debugged, and resolved complex RF issues, collaborating with research and lab teams to improve signal integrity and terminal performance.
- Built automated health monitoring tools, HITL integration testing, CICD pipelines, improving visibility into system-wide performance bottlenecks.

Espresso

Remote

Founder

2021 - 2024

- Developed a B2B SaaS product focused on employee engagement and retention.
- Utilized Slack API and webhooks to allow for users to opt into randomized weekly 1:1 syncs.
- Handled sales/outreach, product, and development as solo founder.
- Product was used by multiple early to mid-stage startups like Bird and Shef.

Palantir

Palo Alto, California

Software Engineer - Backend

2020 - 2023

- Designed and developed multiple backend features for Palantir's collaborative mapping application, Gaia, used for mission planning and operations worldwide.
- Implemented variable grid resolution to improve terrain analysis speed over large continent scale areas by a factor of 10x, improving wait times from minutes to seconds.
- Actively contributed to engineering team culture outside of core responsibilities by interviewing candidates, mentoring interns and new hires, conducting code reviews, and presenting tech talks and engineering reviews.

Tencent Gaming - Lightspeed Studios

Los Angeles, California

Software Engineering Intern

June - December 2018

- Designed and built prototypes for mobile games in Unity that were later pitched to the development team.
- Implemented feature improvements, documented playtesting feedback, and provided technical translations as a contractor at Riot Games for the mobile port of League of Legends, Wild Rift.

EDUCATION

University of California, San Diego

2016 - 2020

Bachelor of Science in Machine Learning; Minor in Computer Science

SKILLS

Programming Languages: Java, C#, Python, C++, C, Rust, JavaScript, SQL, HTML, XML, CSS

Libraries/Engines: Tensorflow, Numpy, Pytorch, ReactJS, Unity, UnrealEngine 5

Language: English (Native), Chinese (Proficient)